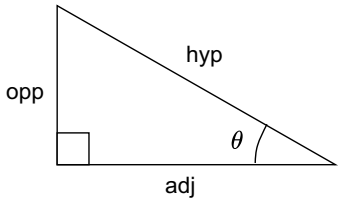
1.39 Trigonometry: Introduction

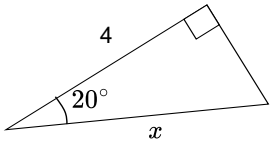
Chapter 9 · Original slide 4

RECAP: Right-Angled Trigonometry

4

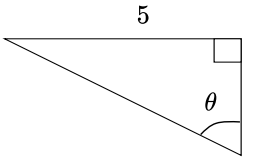


You are probably familiar with the formula: $\sin(\theta) = \frac{opp}{hyp}$
But what is the *conceptual* definition of $\sin$?
sin is a function which inputs an angle and gives the



Find x.

cc
x =



Find θ.

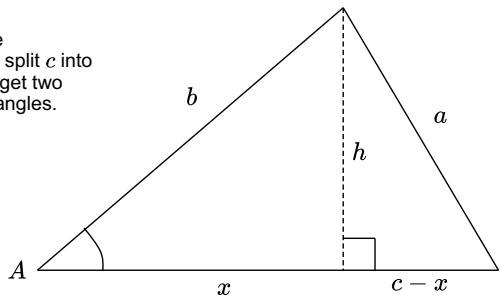
tan
θ =

Chapter 9 · Original slide 5

Proof of Cosine Rule

5

We want to use Pythagoras, so split c into two so that we get two right-angled triangles.



By Pythagoras: $h^2 + x^2 = b^2$ and $h^2 + (c-x)^2 = a^2$
Subtracting to eliminate h:
$x^2 - (c-x)^2 = b^2 - a^2$
$a^2 = b^2 + c^2 - 2cx$
But $x = b\cos A$ ∴
$$a^2 = b^2 + c^2 - 2bc\cos A$$

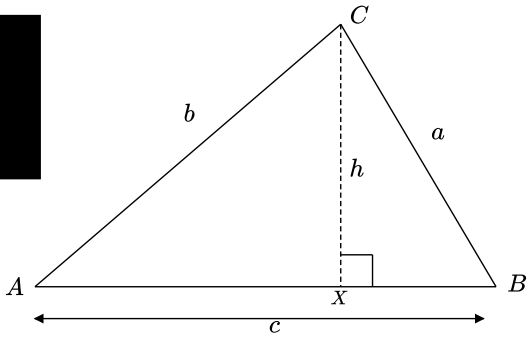
1.39 Trigonometry: Introduction

Chapter 9 · Original slide 6

Proof of Sine Rule

6

The idea is that we can use the common length of ΔACX and $\angle XBC$, i.e. h, to connect the two triangles, and therefore connect their angles/length.



Using basic trigonometry:

$h = a\sin B$ and $h = b\sin A$

$\therefore a\sin B = b\sin A$

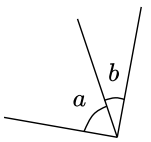
$\therefore \frac{a}{\sin A} = \frac{b}{\sin B}$

Chapter 9 · Original slide 7

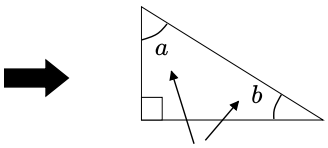
Just for your interest…

7

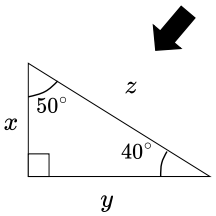
Why does “cosine” contains the word “sine”?



Complementary Angles add to 90°



Therefore these angles are complementary.



$\cos(50) =$ ____

$\sin(40) =$ ____

1.39 Trigonometry: Introduction

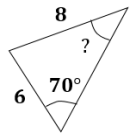
Chapter 9 · Original slide 8

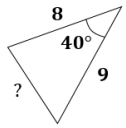
OVERVIEW: Finding missing sides and angles

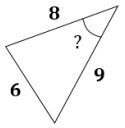
8

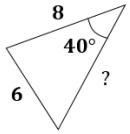
When triangles are not right-angled, we can no longer use simple trigonometric ratios, and must use the cosine and sine rules.

You have	You want
#1: Two angle-side opposite pairs	Missing angle or side in one pair
#2 Two sides known and a missing side opposite a known angle	Remaining side
#3 All three sides	An angle
#4 Two sides known and a missing side not opposite known angle	Remaining side







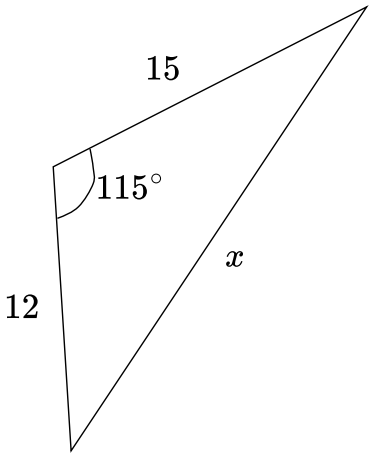


Chapter 9 · Original slide 9

Cosine Rule

9

We use the **cosine rule** whenever we have **three sides** (and an angle) involved.



Proof at end of PowerPoint.

Cosine Rule:

$$a^2 = b^2 + c^2 - 2bc\cos A$$

1.39 Trigonometry: Introduction

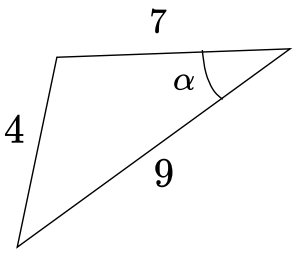
Chapter 9 · Original slide 10

Dealing with Missing Angles

10

You have	You want	Use
All three sides	An angle	Cosine rule

$$\mathbf{a^2 = b^2 + c^2 - 2bc\cos A}$$



Label sides then substitute into formula.

Simplify each bit of formula.

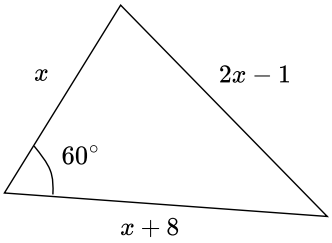
Rearrange "swapping" thing you're subtracting and result

Chapter 9 · Original slide 11

Harder Ones

11

Determine the value of x.



$(2$
$4x$
$3x$
$x^$
$(x$
$\therefore$

[From textbook] Coastguard station B is 8 km, on a bearing of 060° A C

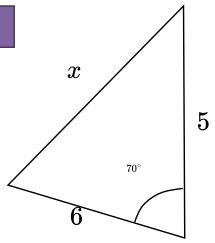
1.39 Trigonometry: Introduction

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Test Your Understanding

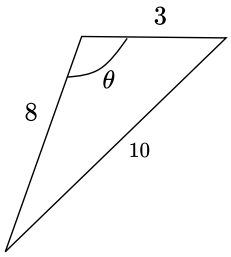
12

1



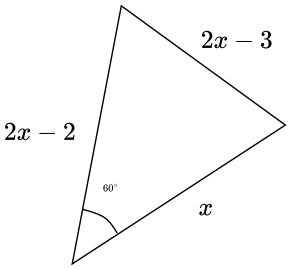
$x =$

2



$\theta =$

3



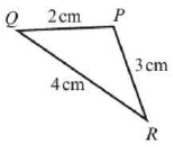
$x =$

Chapter 9 · Original slide 13

Exercise 9A – Y1 pages 177-179

13

8 Show that $\cos P = -\frac{1}{4}$.



11 The lengths of the sides of a triangle are in the ratio $2 : 3 : 4$.
Calculate the size of the largest angle.

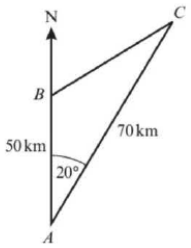
1.39 Trigonometry: Introduction

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Exercise 9A – Y1 pages 177-179

14

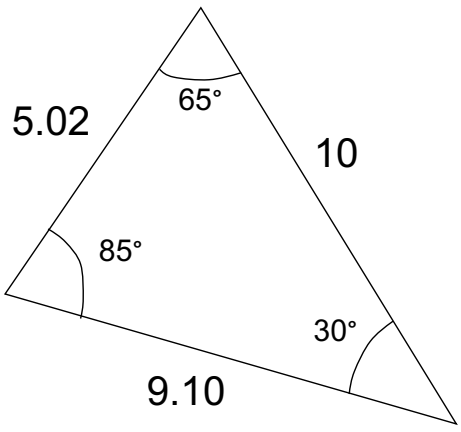
17 The diagram shows three cargo ships, A, B and C, which are in the same horizontal plane. Ship B is 50 km due north of ship A and ship C is 70 km from ship A. The bearing of C from A is $020°$.
a Calculate the distance between ships B and C, in kilometres to 3 s.f. (3 marks)
b Calculate the bearing of ship C from ship B. (4 marks)



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The Sine Rule

16



For this triangle, try calculating each side divided by the sin of its opposite angle. What do you notice in all three cases?

You have	You want	Use
#1. Two angle-side opposite pairs	Missing angle or side in one pair	Sine rule

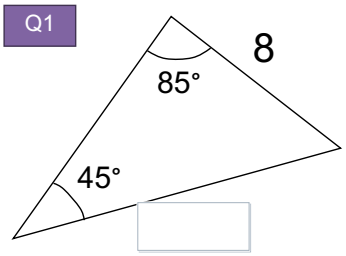
1.39 Trigonometry: Introduction

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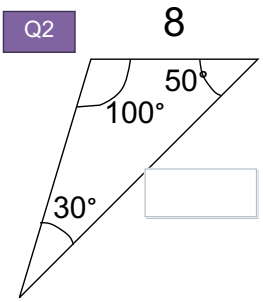
Examples

17

Q1



Q2



You have	You want	Use
#1. Two angle-side opposite pairs	Missing angle or side in one pair	Sine rule

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Examples

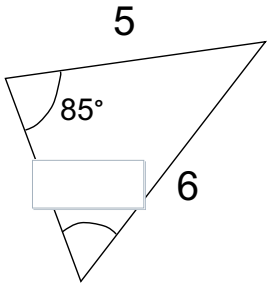
18

When you have a missing angle, it's better to reciprocate to get:

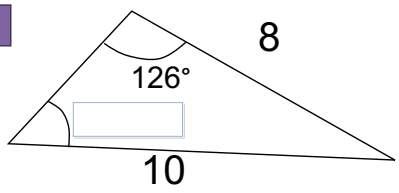
$$\frac{\sin A}{a} = \frac{\sin B}{b}$$

i.e. in general put the missing value in the numerator.

Q3



Q4



1.39 Trigonometry: Introduction

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Exercise 9B – Y1 pages 181-183

19

8 Town B is 6 km, on a bearing of 020°, from town A. Town C is located on a bearing of 055° from town A and on a bearing of 120° from town B. Work out the distance of town C from:
a town A\qquad b town B

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Exercise 9B – Y1 pages 181-183

20

11 In $\triangle ABC$,
$AB = x$ cm, $BC = (4 - x)$ cm, $\angle BAC = y$ and
$\angle BCA = 30^\circ$.
Given that $\sin y = \frac{1}{\sqrt{2}}$, show that

$$x = 4(\sqrt{2} - 1)$$

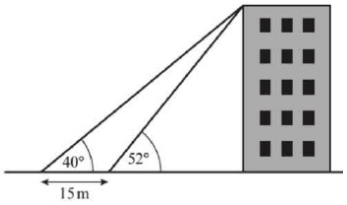
1.39 Trigonometry: Introduction

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Exercise 9B – Y1 pages 181-183

21

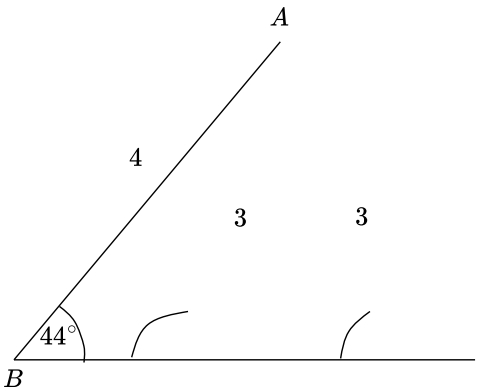
12 A surveyor wants to determine the height of a building. She measures the angle of elevation of the top of the building at two points 15 m apart on the ground.
a Use this information to determine the height of the building. (4 marks)
b State one assumption made by the surveyor in using this mathematical model. (1 mark)



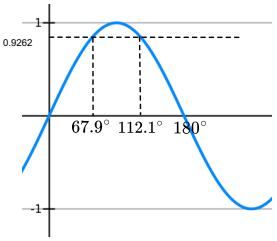
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The ‘Ambiguous Case’

23



Suppose you are told that $AB = 4$, $AC = 3$ and $\angle ABC = 44^\circ$. What are the possible values of $\angle ACB$?



Your calculator will give the acute angle of 67.9° (i.e. C_2). But if we look at a graph of sin, we can see there’s actually a second value for $\sin^{-1}(0.9262)$, corresponding to angle C_1.

The sine rule produces two possible solutions for a missing angle:

$$\sin\theta = \sin(180^\circ - \theta)$$

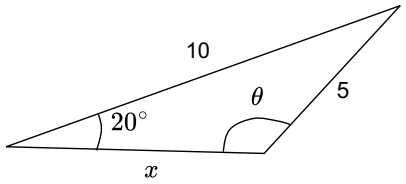
Whether we use the acute or obtuse angle depends on context.

1.39 Trigonometry: Introduction

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Test Your Understanding

24



Given that the angle θ is obtuse, determine θ and hence determine the length of x.

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Exercise 9C – Y1 pages 184-185

25

4 Triangle ABC is such that $AB = 4$ cm, $BC = 6$ cm and $\angle ACB = 36°$. Show that one of the possible values of $\angle ABC$ is $25.8°$ (to 3 s.f.). Using this value, calculate the length of AC.

1.39 Trigonometry: Introduction

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Exercise 9C – Y1 pages 184-185

26

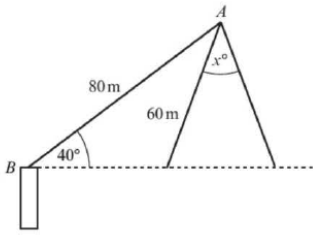
5 Two triangles ABC are such that $AB = 4.5\text{ cm}$, $BC = 6.8\text{ cm}$ and $\angle ACB = 30°$. Work out the value of the largest angle in each of the triangles.

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Exercise 9C – Y1 pages 184-185

27

6 a A crane arm AB of length 80 m is anchored at point B at an angle of $40°$ to the horizontal. A wrecking ball is suspended on a cable of length 60 m from A. Find the angle x through which the wrecking ball rotates as it passes the two points level with the base of the crane arm at B. (6 marks)
b Write down one modelling assumption you have made. (1 mark)

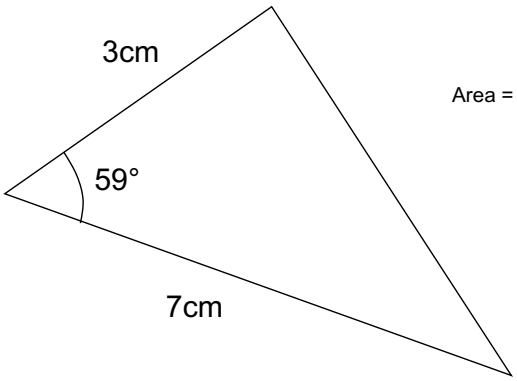


1.39 Trigonometry: Introduction

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Area of Non Right-Angled Triangles

28



Area =

$$\text{Area} = \frac{1}{2}ab\sin(C)$$

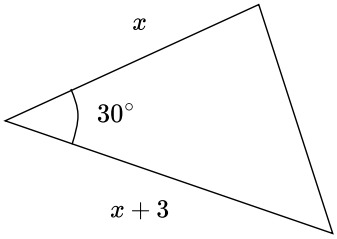
where C is the angle between two sides a and b.

Tip: You shouldn't have to label sides/angles before using the formula. Just remember that the angle is between the two sides.

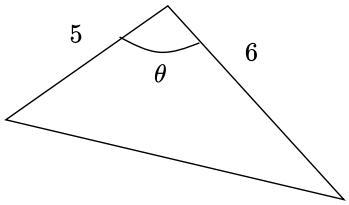
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Test Your Understanding

29



The area of this triangle is 10.
Determine x.



The area of this triangle is also 10.
If θ is obtuse, determine θ.

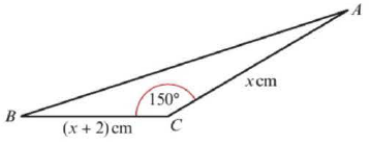
1.39 Trigonometry: Introduction

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Exercise 9D – Y1 pages 186-187

30

4 In triangle ABC, $BC = (x + 2)$ cm, $AC = x$ cm and $\angle BCA = 150^\circ$. Given that the area of the triangle is 5 cm^2, work out the value of x, giving your answer to 3 significant figures.



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Exercise 9D – Y1 pages 186-187

31

5 In $\triangle PQR$, $PQ = (x + 2)$ cm, $PR = (5 - x)$ cm and $\angle QPR = 30^\circ$. The area of the triangle is A cm^2.

a Show that $A = \frac{1}{4}(10 + 3x - x^2)$.

b Use the method of completing the square, or otherwise, to find the maximum value of A, and give the corresponding value of x.

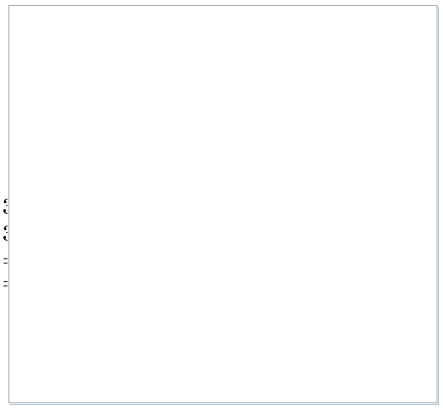
1.39 Trigonometry: Introduction

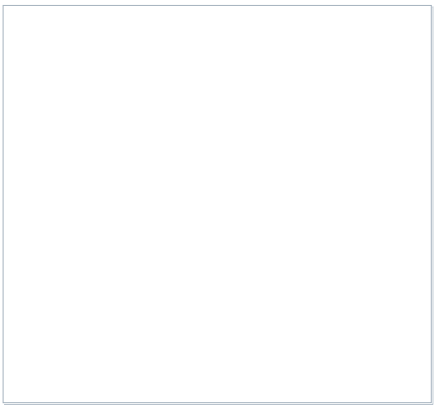
Chapter 9 · Original slide 32

Exercise 9D – Y1 pages 186-187

32

6 In $\triangle ABC$, $AB = x$ cm, $AC = (5 + x)$ cm and $\angle BA$
that the area of the triangle is $3\frac{3}{4}$ cm^2
a Show that x satisfies the equation $x^2 + 5x - 15 = 0$.
b Calculate the value of x, giving your answer to 3 significa... figures.





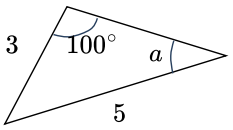
Chapter 9 · Original slide 33

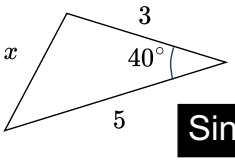
Sin or cosine rule?

33

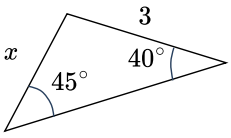
Recall that whenever we have **two "side-angle pairs"** involved, use sine rule. If there's **3 sides** involved, we can use cosine rule. Sine rule is generally easier to use than cosine rule.

Sine Cosine

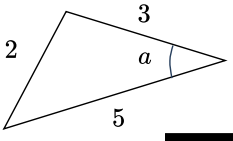




Sine Cosine



Sine Cosine



Sine Cosine

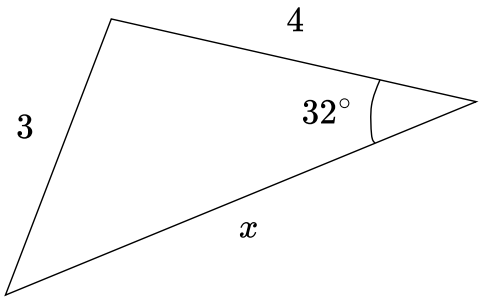
1.39 Trigonometry: Introduction

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Using sine rule twice

34

You have	You want	Use
#4 Two sides known and a missing side not opposite known angle	Remaining side	Sine rule twice



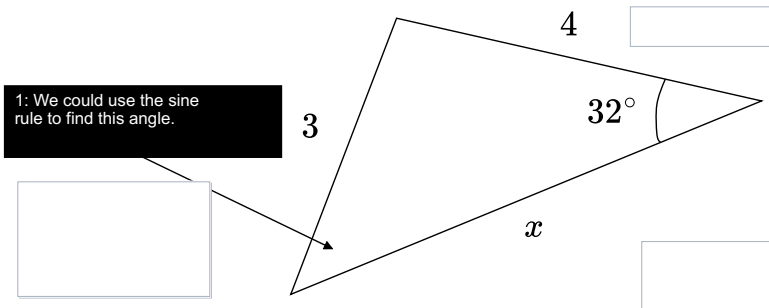
Given there is just one angle involved, you might attempt to use the cosine rule:

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Using sine rule twice

35

You have	You want	Use
#4 Two sides known and a missing side not opposite known angle	Remaining side	Sine rule twice



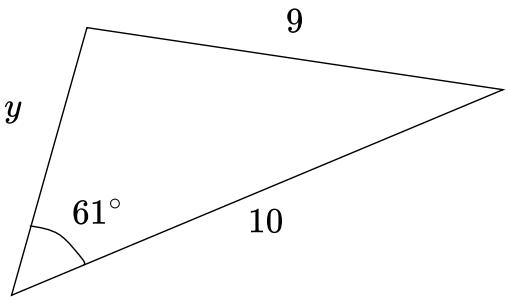
1: We could use the sine rule to find this angle.

1.39 Trigonometry: Introduction

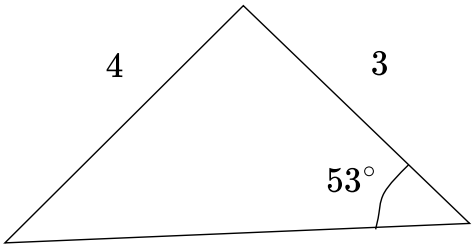
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Test Your Understanding

36



$y =$



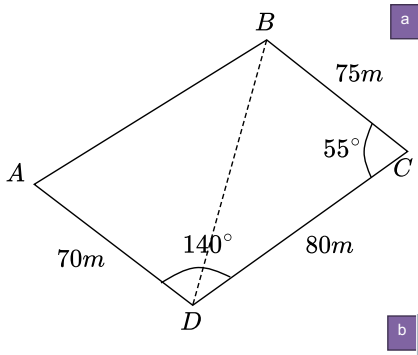
$Area =$

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Problem Solving With Sine/Cosine Rule

37

[From Textbook] The diagram shows the locations of four mobile phone masts in a field, $BC = 75m$. $CD = 80m$, angle $BCD = 55°$ and angle $ADC = 140°$.
In order that the masts do not interfere with each other, they must be at least 70m apart.
Given that A is the minimum distance from D, find:
a) The distance A is from B
b) The angle BAD
c) The area enclosed by the four masts.



a

b

c

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Exercise 9E – Y1 page 189-191

38

12 The perimeter of $\triangle ABC = 15\ \text{cm}$. Given that $AB = 7\ \text{cm}$ and $\angle BAC = 60°$, find the lengths of AC and BC and the area of the triangle.

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Exercise 9E – Y1 page 189-191

39

13 In the triangle ABC, $AB = 14\ \text{cm}$, $BC = 12\ \text{cm}$ and $CA = 15\ \text{cm}$.
a Find the size of angle C, giving your answer to 3 s.f.
b Find the area of triangle ABC, giving your answer in cm^2 to 3 s.f.

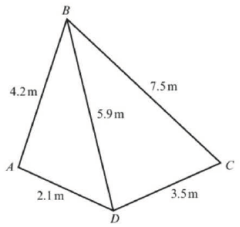
1.39 Trigonometry: Introduction

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Exercise 9E – Y1 page 189-191

40

14 A flower bed is in the shape of a quadrilateral as shown in the diagram.
a Find the sizes of angles DAB and BCD. \hfill(4 ma...
b Find the total area of the flower bed. \hfill(3 ma...
c Find the length of the diagonal AC.



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Exercise 9E – Y1 page 189-191

41

15 $ABCD$ is a square. Angle CED is obtuse.
Find the area of the shaded triangle. (7 marks)

